

charged particles generated in solar flares, which travel into interplanetary space. Other cosmic occurrences also involve energy-release processes that are similar. These occurrences take place on distant objects that cannot be investigated in as much depth as solar flares on the Sun. Understanding solar flares can help us better comprehend these occurrences. Solar flares offer a chance to explore natural physical processes that resemble those that take place in lab equipment designed with the goal of establishing controlled thermonuclear fusion.

Importance of Heliophysics

NASA and other space agencies are studying the Sun and how it affects the essential nature of space, which in-turn affects planet atmospheres and technology. Contrary to popular belief, space is not entirely empty; rather, we reside in the prolonged atmosphere of a burning star. The solar wind, which is produced by our Sun, is a continuous stream of particles and energy, and its magnetic field is also ever-writhing. The Sun, Earth, and the planets are all surrounded by this vast, active solar atmosphere, which also reaches far beyond the Solar System. Researching this structure not only helps in our comprehension of the fundamental principles governing the cosmos, but it also serves to safeguard our technology and space personnel. Because extreme space weather can disrupt our communications, satellites, and power grids, NASA researches the region around the Earth. We can learn further about how stars affect the habitation of planets across the universe by studying the Sun and space.

It takes a comprehensive investigation of a massive system dispersed across the Solar System to map out this integrated structure. From the Parker Solar Probe, which observes the very beginning of solar wind, to satellites orbiting

the Earth, and the farthest object constructed by humans, Voyager, NASA has an assortment of spacecraft strategically positioned across our heliosphere.

How are Solar Storms observed?

It started with Dr. Don Gurnett, a space physicist and professor at the University of Iowa, who captured the vibrations of two of the most powerful solar flares recently, as they traveled away from the Sun using NASA's Cassini spacecraft. The solar flare visible in the bottom center of the image was photographed by NASA's Solar and Heliospheric Observatory (SOHO) spacecraft early on October 28, 2003.

The radio wave burst was captured by Cassini on October 28 while it was on its route to its rendezvous with Saturn on July 1, 2004. It resembled the clicking of an antiquated telegraph machine accompanied by the frantic roar of a jet engine. The first spacecraft to provide a close-up view of the ringed planet and its moons will be Cassini. The European Space Agency's tacked-on Huygens probe will descend through Titan's smoggy atmosphere and make contact with what may or may not be a fluid methane ocean. The Solar and Heliospheric Observatory (SOHO) project, a collaboration between NASA and ESA, was created to investigate the Sun from the inside out, including its internal composition, vast outer atmosphere, and solar wind that it blasts throughout the solar system.

Completed in December 1995, SOHO was intended to run until 1998, but due to its effectiveness, NASA and the European Space Agency have approved multiple mission extensions over the following 20 years, enabling it to observe many solar cycles or 11-year phases of solar activity. In the last two decades, the convection zone - the Sun's turbulent uppermost layer of interior - and the framework of sunspots beneath the surface have been captured for the



Source: NASA

SOHO captured this image on
Oct 28, 2003

first time by SOHO. In-depth observations of the interior, the slow and rapid solar wind, event identification source regions, and acceleration mechanisms have all been produced by it. Additionally, the mission has uncovered more than 3,000 brand-new comets as well as brand-new dynamic solar phenomena like coronal waves and solar storms.

Twelve instruments created by 29 distinct institutes in 15 countries are available at SOHO. The extensive range of instruments allows SOHO to conduct a wide range of scientific investigations, including imaging the Sun and its atmosphere, examining the different light wavelengths and their makeup, and monitoring energetic particles that pass the spacecraft. Since 2010, the NASA Solar Dynamics Observatory (SDO) has been observing the Sun with a full complement of equipment. To gather information on different solar activities, it makes use of three highly important pieces of equipment. They include the Extreme Ultraviolet Variability Experiment (EVE), which measures the Sun's extreme ultraviolet irradiance; the Helioseismic and Magnetic Imager (HMI), which performs high-resolution metrics of the longitudinal and vector magnetic field over the entire visible solar disc; and the Atmospheric Imaging Assembly (AIA), which offers constant observations of the solar chromosphere and corona in seven extreme ultraviolet (EUV) channels. **d**